

Anhydrous proton conduction at 150 °C in a crystalline metal-organic framework

Jeff A. Hurd¹, Ramanathan Vaidhyanathan¹, Venkataraman Thangadurai¹, Christopher I. Ratcliffe², Igor L. Moudrakovski² and George K. H. Shimizu^{1*}

Metal organic frameworks (MOFs) are particularly exciting materials that couple porosity, diversity and crystallinity. But although they have been investigated for a wide range of applications, MOF chemistry focuses almost exclusively on properties intrinsic to the empty frameworks; the use of guest molecules to control functions has been essentially unexamined. Here we report Na₃(2,4,6-trihydroxy-1,3,5-benzenetrisulfonate) (named β -PCMOF2), a MOF that conducts protons in regular one-dimensional pores lined with sulfonate groups. Proton conduction in β -PCMOF2 was modulated by the controlled loading of 1*H*-1,2,4-triazole (Tz) guests within the pores and reached $5 \times 10^{-4} \text{ S cm}^{-1}$ at 150 °C in anhydrous H₂, as confirmed by electrical measurements in H₂ and D₂, and by solid-state NMR spectroscopy. To confirm its potential as a gas separator membrane, the partially loaded MOF (β -PCMOF2(Tz)_{0.45}) was also incorporated into a H₂/air membrane electrode assembly. The resulting membrane proved to be gas tight, and gave an open circuit voltage of 1.18 V at 100 °C.

Metal organic frameworks (MOFs) are breakthrough materials characterized by tunable pores permeating regular molecular scaffoldings^{1,2}. These inorganic solids couple the variety of potential ligands from coordination chemistry with the topological diversity of solid-state chemistry. To date, most research on MOFs has focused on gas storage applications³, with particular interest for hydrogen^{4,5}, carbon dioxide^{6,7} and methane⁸. These solids are noted for their crystallinity, which enables molecular-level structure determination by X-ray techniques. Transport phenomena, such as ion conduction, are more widely associated with disordered structures as movement of components in a solid typically disfavors crystallinity and, to date, studies on transport properties in crystalline MOF-type materials are rare^{9–11}.

A phenomenon increasingly evidenced in coordination polymer materials is their ability to undergo reversible structural changes in the solid state in response to guest molecules, temperature or pressure^{12–15}. For porous network solids, this point is itself remarkable as the solid must also be sufficiently stable to sustain pores. MOFs have shown the ability to reconcile stability and structural dynamics, and often achieve this with retention of crystallinity. Our team has been pursuing organosulfonate MOFs as the sulfonate group imparts ‘adaptable’ coordinative properties¹⁶ that are often manifested as reversible solid-state dynamics. Given that, and the well-established role of macromolecular sulfonic acid polymers as proton conductors^{17–20}, we undertook a study of sulfonated MOFs as proton-conducting solids.

From the perspective of synthesizing new materials, there are several potential advantages to this approach. The open framework structure of MOFs could allow for controlled guest loading with non-volatile proton carriers, enabling proton conduction over 100 °C. Being in the microporous realm, the open structure could be blocked with guest molecules to prevent hydrogen fuel crossover in a fuel cell configuration. From a design perspective, owing to the crystallinity of MOFs, molecular level insights into transport phenomena could be gained. Coupling this with the modularity of

MOF synthesis could open the door to a wide range of potentially better ion conductors. Here, we report a crystalline sulfonate coordination material, β -PCMOF2 (proton conducting metal organic framework 2) with channels that can be loaded with amphoteric heterocycles to exhibit proton conduction under anhydrous H₂ at 150 °C. This is a new domain for MOF materials and also a different approach to proton-conducting solids²¹. The results are enabled by not only the controlled loading of certain guests in a MOF, but also by the specific stoichiometries of that guest. As a step towards application, we show that loaded β -PCMOF2 can function both as a proton conductor and gas separator in a membrane electrode assembly (MEA) and exhibit open circuit voltages (OCVs) over 1.1 V at 100 °C for a H₂-air fuel cell. This is significant as membrane materials that function above the boiling point of water are needed for widespread use of proton exchange membrane (PEM) and direct methanol fuel cells (DMFCs).

Results

The trisodium salt of 2,4,6-trihydroxy-1,3,5-benzenetrisulfonate, L, can be prepared as a lower-temperature α -phase (not discussed here) and a higher-temperature β -phase. β -PCMOF2, crystallizes in a honeycomb structure with one-dimensional pores lined with sulfonate groups. The L molecules are disordered because of rotation in the plane of the benzene ring; each part of Fig. 1 represents one position of the molecule. The X-ray structure shows hexagonal sheets of L molecules, in the crystallographic *ab* plane, crosslinked in the third dimension by Na ions to form channels (Fig. 1a). The channels in β -PCMOF2 have an 18-membered periphery with a diameter of 5.65(2)–5.91(2) Å measured between Na ions and the transannular O atoms. The Na⁺ ions are eight-coordinate and ligated by two hydroxyl (Na1–O6 = 2.669(2)–2.676(2) Å) and six sulfonate oxygen atoms (Na1–O = 2.350(2)–2.798(2) Å). The sheets of ligands are packed efficiently with the nearest intermolecular C–C distance at 3.987(2) Å (Fig. 1b). Significantly, the pores are lined exclusively with oxygen atoms of sulfonate groups (Fig. 1c,d) to form a potential hydrogen-bond transfer pathway. Figure 2 shows powder X-ray

¹Department of Chemistry, University of Calgary, Calgary, Alberta, T2N 1N4, Canada, ²Steele Institute for Molecular Sciences, National Research Council, 100 Sussex Drive, Ottawa, Ontario K1A 0R6, Canada. *e-mail: gshimizu@ucalgary.ca

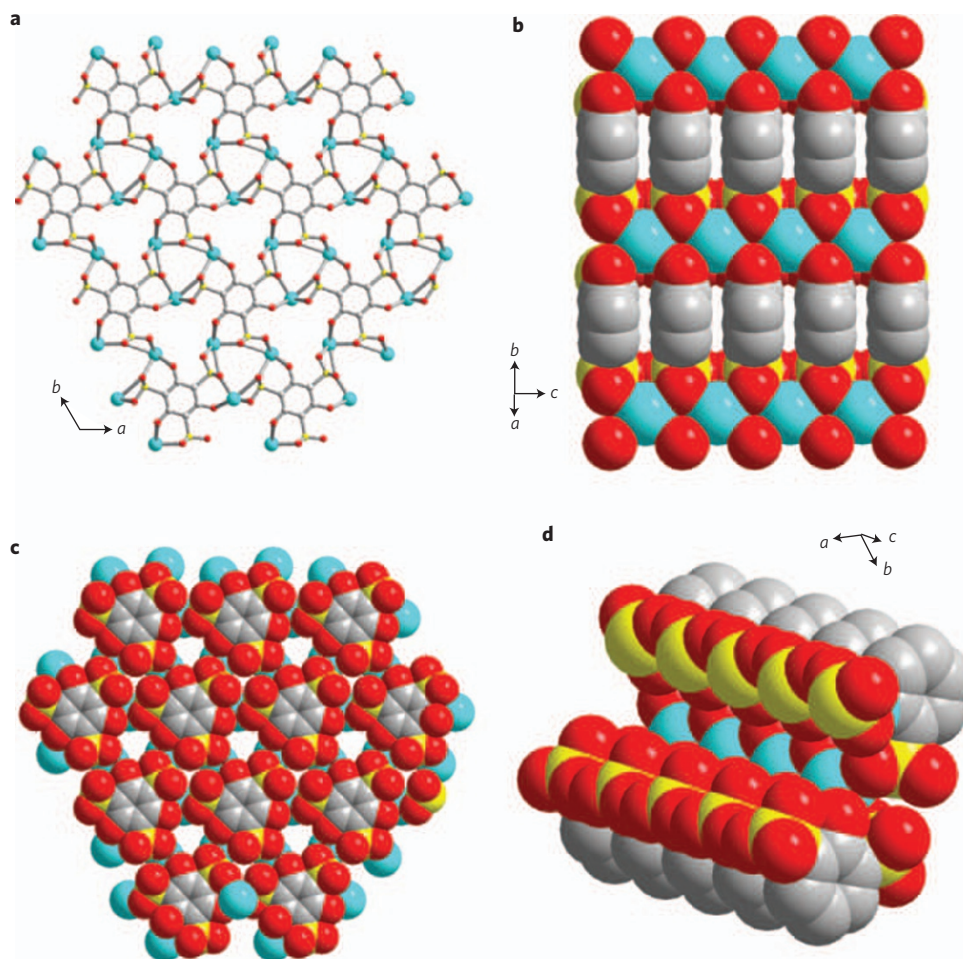


Figure 1 | Single-crystal X-ray structure of β -PCMOF2. Grey, C; red, O; yellow, S; cyan, Na. **a**, A ball-and-stick representation of a single layer showing the open 18-membered channels. **b**, A space-filling diagram perpendicular to the pores, showing the packing of the aryl groups down the crystallographic c -axis. **c**, A space-filling diagram down the axis of the pores. **d**, A space-filling diagram of the cross-section of one pore, showing the high degree of sulfonation.

data confirming that β -PCMOF2 can be prepared as a matching bulk sample.

Solid-state ion conduction involves passage of an ion between favourable binding sites. In polymers such as Nafion, sulfonic acid aggregates act as scaffolds for water-mediated proton transfer^{18–20}. To assess whether the ‘persulfonated’ pore wall structure in β -PCMOF2 could conduct protons, two-probe a.c. impedance spectroscopy was performed (Supplementary Figs S10–S13). Experiments were performed using porous carbon electrodes, anhydrous hydrogen gas and varying temperature and humidity. The impedance plots of β -PCMOF2 at 150 °C were resolved using the equivalent circuit $(R_bCPE_b)(R_{gb}CPE_{gb})(CPE_{el})$ where R_b = resistance inherent to the bulk material; R_{gb} = resistance attributed to interfacial grain-boundary contributions; CPE = contributions to account for the depressed, versus perfect, semicircle due to non-ideal capacitance—so called constant phase elements—for bulk (b), grain boundary (gb) and electrode (el) contributions. The impedance plots were reproducible during heating and cooling cycles. The CPE_b value was of the order 10^{-11} F at high frequency and about 10^{-9} F at low frequency regions of the semicircle, typical for bulk and grain-boundary contributions. The spike at low frequency is attributed to electrode polarization. At 30 °C, β -PCMOF2(H_2O)_{0.5} conducts protons at 5.0×10^{-6} S cm⁻¹ (Fig. 3). Heating above 70 °C causes dehydration and a decrease in conduction to below 10^{-8} S cm⁻¹. To examine hydration dependence, the sample was humidified prior to the analysis of β -PCMOF2(H_2O)_{0.6}. Over the temperature range analysed, the conductivity was raised by

0.5–1.0 orders of magnitude, but above 70 °C conductivity still dropped below 10^{-7} S cm⁻¹. Loss of proton conduction with dehydration is typical for sulfonated ionomers because, for Grotthuss or vehicle mechanisms, H_2O molecules are critical^{18–20}.

For PEMs and direct methanol fuel cells, a major challenge is proton conduction above 100 °C (see below)^{17–23}. An approach to attain this goal is to replace water as the proton carrier with an amphoteric N -heterocycle, such as imidazole^{24,25}. The imidazole/imidazolium pair can transfer protons via a Grotthuss-type mechanism and are also a less volatile base–acid pair. As the channels in β -PCMOF2, were sufficiently large to include such heterocycles, this approach was examined. Our results with 1*H*-1,2,4-triazole (Tz) are presented here. In our studies, Tz was found to be a superior anhydrous proton transfer agent than imidazole, as has been put forth by Liu²⁶.

Uniform loading of one-dimensional channels is a challenge as the core regions of individual particles are often inaccessible²⁷. Incorporation of the heterocycle into the channels of β -PCMOF2 was achieved by *in situ* loading of Tz into the α -phase and conversion to the β -phase with heating as confirmed by several techniques (see Supplementary Information). Powder X-ray diffraction (PXRD) showed that the Tz-loaded networks were isostructural to the unloaded hydrated β -phase (Fig. 2) and ¹³C and ¹H solid-state NMR (Supplementary Figs S17–S22) confirmed framework retention and the absence of free triazole. Samples of β -PCMOF2(Tz)_{*x*} were prepared with various controlled loadings ($x = 0.3, 0.45, 0.6$).

Typical impedance plots of hydrated and Tz-loaded β -PCMOF2, obtained using porous carbon electrodes at 150 °C, show at low

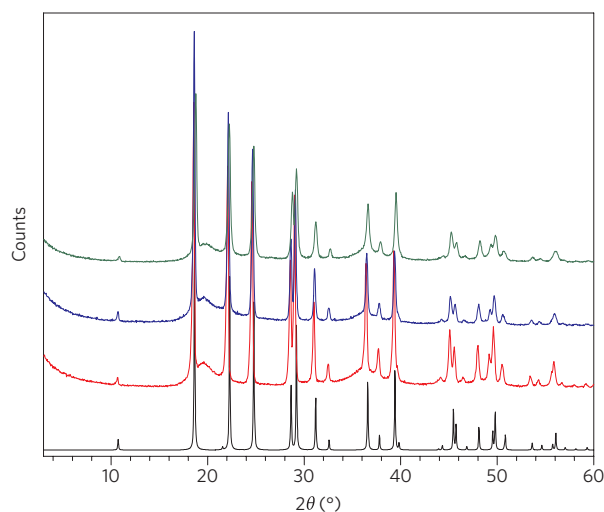


Figure 2 | Comparison between the simulated and experimental powder X-ray diffraction (PXRD) patterns for β -PCMOF2. Black: simulation from the single-crystal structure of β -PCMOF2. Red: a bulk preparation of β -PCMOF2. Blue: a bulk preparation of β -PCMOF2(Tz)_{0.45}. The patterns for the $x = 0.3$ and 0.6 loadings are analogous. Green: a sample of a β -PCMOF2(Tz)_{0.45} run after impedance experiments with two cycles of heating to 150°C . There is a slight shift to higher 2θ values, which is attributed to dehydration of the sample in the first run.

temperatures a semicircle in the high frequency region and a tail at low frequencies (Supplementary Figs S11–S14). The tail at low frequencies may be due to mobile ions being blocked by the electrode–electrolyte interfaces and/or instrument limitations in observing the low-frequency electrode polarization region. Owing to large overlaps at high temperature, the bulk, grain-boundary and electrode contributions were unable to be separated. Therefore, the low-frequency minimum/intercept along the Z' -axis of the impedance plots was used to estimate bulk resistance. Accordingly, the reported calculated electrical conductivity may be considered as the lower limit of the actual bulk conductivity of these PCMOFs. Samples were run through two cycles of heating to 150°C (holding for 12 h) and were then cooled to 23°C in anhydrous H_2 with excellent reproducibility. Comparable a.c. impedance plots are observed for a wide range of known proton conductors²⁸ including Nafion membranes^{18–20,29}, CsHSO_4 (ref. 30), phosphotungstic acid/polystyrenesulfonic acid composites³¹, uracil alkylphosphate composites³², and Ta-doped $\text{Ba}_3(\text{CaNb})_3\text{O}_9$ double perovskites³³.

Impedance analyses show that the loaded β -PCMOF2(Tz) _{x} samples are much better proton conductors than their unloaded, hydrated counterparts, and indeed function above 100°C . The data also confirm a dependence of proton conductivity on Tz loading. Figure 3 shows Arrhenius plots from the impedance analysis of four samples of β -PCMOF2(Tz) _{x} ($x = 0, 0.30, 0.45$ and 0.60). At 150°C , the conductivity measured for $x = 0.30, 0.45$ and 0.60 Tz-loaded samples reached roughly $2 \times 10^{-4} \text{ S cm}^{-1}$, $5 \times 10^{-4} \text{ S cm}^{-1}$, and $4 \times 10^{-4} \text{ S cm}^{-1}$, respectively. Neither pure triazole nor pure β -PCMOF2 show proton conductivity above $10^{-6} \text{ S cm}^{-1}$. PXRD analyses confirmed the retention of the original structure after impedance analyses (Fig. 2).

In order to confirm that the observed electrical effects were due to proton conduction, and not electron or Na^+ conduction, a.c. impedance experiments were performed in H_2 and D_2 , with and without gas diffusion electrodes. In Fig. 4, the a.c. impedance plots of PCMOF-2(Tz)_{0.3} in anhydrous H_2 and D_2 , with gas diffusion electrodes, are shown. Interestingly, three clear semicircles due to bulk, grain-boundary and electrode polarization contributions were observed in both H_2 and D_2 . A noteworthy feature is the

absence of the tail at low frequencies indicating the non-blocking nature of the electrolyte–electrode interfaces. Also, for D_2 , the expected kinetic isotope effect was observed in that the bulk resistance was ~ 1.5 times higher than H_2 , which is consistent with deuteron versus proton migration³³.

For the β -phase, in the bulk PXRD (Fig. 2), a broad ‘hump’ centered at $\sim 20.5^\circ 2\theta$ was observed. This feature was present both with pure and Tz-loaded β -PCMOF2. Two possible explanations exist. The first is that an amorphous-like impurity was present, which is disconcerting as a less-ordered phase could contribute to ion conduction. The second explanation is that the broad peak arose from the β -PCMOF2 sample itself. As the anomaly was present in pure β -PCMOF2, it could not be attributed to the triazole loading. A key point in ionic conduction measurements is that sufficient electrolyte must be present to form a continuous pathway between electrodes; assuming a random particle distribution, at least half the material should be composed of the conducting phase. The phase purity of β -PCMOF2 was examined by scanning electron microscopy (SEM) and showed a homogeneous sample of plate-like crystallites (Supplementary Figs S23–S26). The exact sample shown in the SEM images in Supplementary Figure S24a,b gave the PXRD pattern shown in Fig. 2. Furthermore, if the ‘hump’ represented a different phase, some change in its relative proportion would be anticipated with different preparative conditions; no such variation was observed through six trials. With the crosslinked layered structure of β -PCMOF2, we believe the broad peak most likely originates from incommensurate lattice packing. β -PCMOF2 does not conduct protons in the absence of triazole even though the broad peak appears in its PXRD pattern.

The activation energy for proton conduction was calculated by fitting the electrical conductivity data to the Arrhenius equation. β -PCMOF2(Tz) _{x} , $x = 0.3$, clearly demonstrated Arrhenius behaviour over the investigated range of 23 – 150°C with an activation energy of 0.51 eV . The $x = 0.45$ and 0.6 samples exhibited a change in slope and two activation energies have been determined. For the $x = 0.45$ sample, the activation energy was 1.8 eV between 50 and 90°C , and 0.34 eV between 90 and 150°C . For the $x = 0.6$ sample, the activation energy was 1.87 eV between 23 and 80°C , and 0.56 eV between 80 and 150°C . Lower activation energies at higher temperatures have also been observed in other solid-state

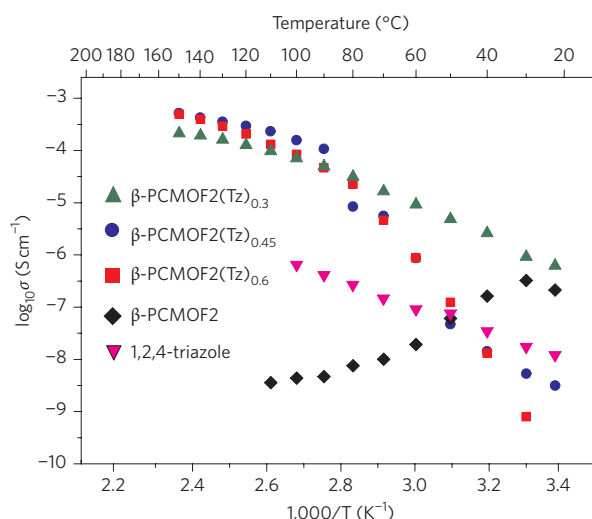


Figure 3 | Comparison of the Arrhenius plots of β -PCMOF2, $[\beta\text{-PCMOF2(Tz)}_{0.3}]$, $[\beta\text{-PCMOF2(Tz)}_{0.45}]$, and $[\beta\text{-PCMOF2(Tz)}_{0.6}]$ measured in anhydrous H_2 atmospheres (data from the second heating cycles is shown to remove hydration). For comparison, the electrical conductivity for pure 1,2,4-triazole is also shown in pink. σ , conductivity.

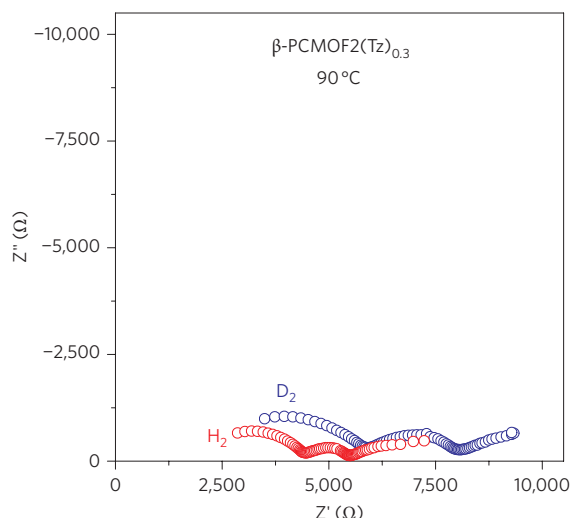


Figure 4 | a.c. impedance plots of as-pressed pellets of β -PCMOF2(Tz)_{0.3} at 90 °C in dry H₂ and D₂. Both curves show clear separation of bulk, grain-boundary and electrode contributions, and the D₂ plot (blue) shows the expected kinetic isotope effect relative to H₂ (red). Measurements were performed using Pt-C gas diffusion electrodes (5 mg Pt m⁻², ELAT, BASF fuel cell).

ion conductors where superionic phase transitions enhance the mobility of ion vacancies³⁰. Mechanistic studies with loading dependence are the subject of ongoing work but lower activation energies at higher loading would be consistent with facilitated hydrogen bond transfer by the heterocycles. Variable temperature ¹H magic angle spinning solid-state NMR studies (Supplementary Figs S19–S22) were performed on hydrated β -PCMOF2, β -PCMOF2(Tz)_x ($x = 0.3, 0.6$) and pure Tz up to 120 °C. These spectra confirm that the PCMOF structure was retained, that the triazole was incorporated (that is, no free Tz present) and that the protonation state of the Tz changed with temperature, all consistent with the proposed model.

As a critical proof-of-principle experiment, to further confirm the potential of a MOF as a membrane, a membrane–electrode assembly with β -PCMOF2(Tz)_{0.45} was constructed and the electromotive force of the H₂/air cell was measured. The MEA was prepared by tying platinum wires to porous Pt-C electrodes. β -PCMOF2(Tz)_{0.45} powder was pressed between the electrodes in a die, with the Pt wires away from the sample. A gas-tight connection, via a hollow ceramic tube, was made between the H₂ source and the anode. The other side of the MEA was left open to air as the cathode. Electromotive force measurements (Fig. 5) showed an open circuit voltage of 1.18 V, stable for 72 h at 100 °C. There was a decrease in the OCV value to 0.93 and 0.77 V at 120 °C and 140 °C, respectively, which was attributed to fuel crossover.

Discussion

In Fig. 6, we compare the proton conduction of [β -PCMOF2(Tz)_{0.3}] with well-known proton conductors reported in the literature, particularly focusing on the region over 100 °C. Interestingly, the Tz-loaded PCMOF exhibits conductivity about four orders of magnitude higher than Zr-phosphate and β -alumina-based ceramic proton conductors²⁴. It is also three orders of magnitude lower than that of fully hydrated Nafion and H₃PO₄-derived proton conductors²⁴. However, the proton conductivity of Nafion and PBI-H₃PO₄ (ref. 34) depend highly on humidity, whereas the loaded PCMOF shows proton conduction in anhydrous conditions. More importantly, one can pose the question for the other materials charted, ‘What are the options for rationally designing improvements?’ From a materials design perspective, the MOF approach

offers a precisely known structure and ample options for systematic, modular variation to improve conductivity and robustness.

The demonstration of anhydrous proton conduction in a MOF is only a step towards applicability as a fuel cell membrane. Anhydrous proton conduction above 100 °C is significant to fuel cell technologies as, for PEMFC and DMFC systems, efficiencies are improved by operation above 100 °C^{17–23}. The reasons are multifold and include: enhancing electrode kinetics; decreasing CO poisoning of Pt-based electrodes; management of only a single phase of water above 100 °C. However, basic issues that could disqualify this electrolyte include: the ability to construct a dense membrane from the powder; inadequate PCMOF/electrode contact; partial electronic conductivity of the MOF; or catastrophic fuel crossover. The open circuit voltage experiment on the PCMOF-electrode assembly shows that none of these potential issues are overriding; a guest-loaded, proton-conducting MOF can function to separate H₂ from the cathode half-cell and sustain a marked potential difference across the electrolytic cell at 100 °C. With respect to fuel crossover, the one-dimensional pores shown in Fig. 1 would appear to invite passage of fuel between half-cells. However, a key point for β -PCMOF2 is that Tz sits in, and is only slightly smaller than, the pores (Tz is 4.21 Å across whereas the pores are 5.6 Å). The value of 1.18 V for an OCV is near the theoretical maximum, at 100 °C and 1 atm pressure, for the formation of water vapour from the H₂–O₂ cell (1.16 V). Although it is higher than the theoretical value, the uncertainty in our OCV measurement itself is ± 0.02 V, so it is within error. Further variation also comes from the temperature fluctuation in the tube furnace. The OCV value of 1.18 V at 100 °C is better than that of low temperature Nafion-based fuel cells³⁵.

To contrast the MOF approach with other crystalline materials, solid acids possess crystalline architectures, which give insight into the mechanism of ion conduction, and have shown very good proton conductivity well above 150 °C (ref. 36). Some of these materials can have both chemical and mechanical stability issues but there are options for varying inorganic components. However,

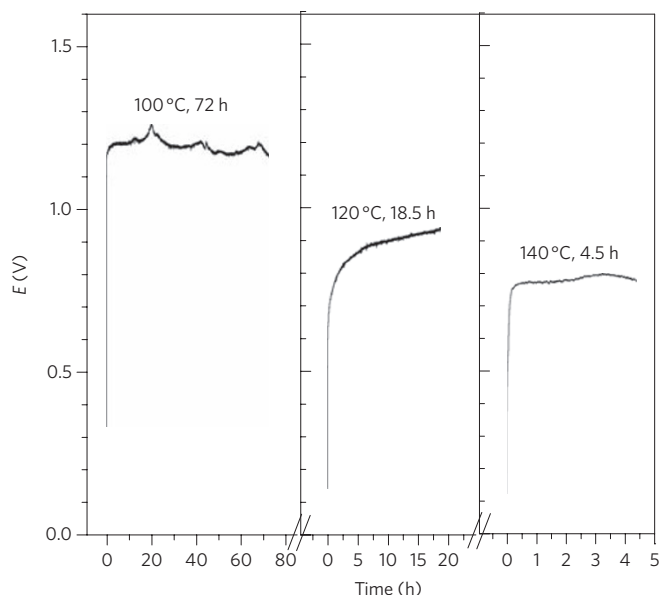


Figure 5 | Open-circuit voltage measurements performed over time at different temperatures, using the following electrochemical cell: H₂, Pt,C| β -PCMOF2(Tz)_{0.45}|Pt,C, air. The relatively constant voltages at the different temperatures demonstrate the stability of the β -PCMOF2(Tz)_{0.45} material to the operating conditions. Note that the sample was equilibrated at each temperature prior to taking a measurement.

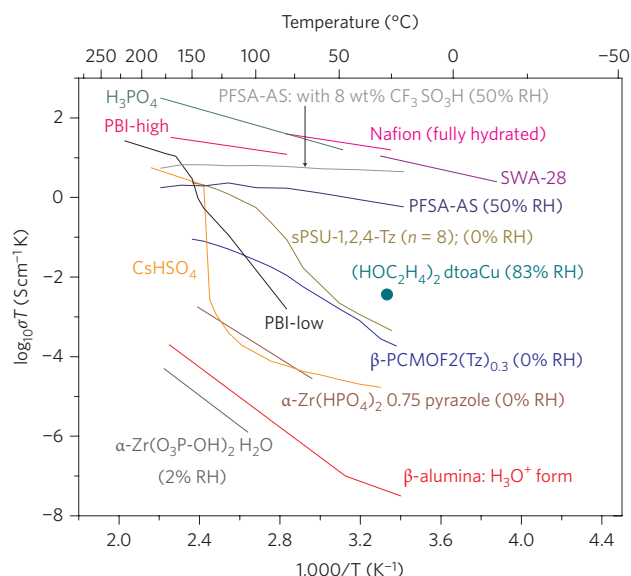


Figure 6 | Comparison of β -PCMOF2(Tz)_{0.3} with other proton-conducting materials. SWA is the highly hydrated silicotungstic acid $\text{H}_4\text{SiW}_{12}\text{O}_{40} \cdot 28\text{H}_2\text{O}$. sPSU-1,2,4-Tz is sulfonated polysulfone with intercalated 1H-1,2,4-triazole (where n refers to the ratio of Tz/ SO_3H). PBI is polybenzimidazole, where high and low refer to the H_3PO_4 content. $(\text{HOC}_2\text{H}_4)_2\text{dtoaCu}$ is N,N' -bis-(2-hydroxyethyl)dithiooxamidatocopper(II), a two-dimensional coordination polymer. PFSA-AS is a mesoporous aluminosilicate with covalently grafted perfluorosulfonic acid groups.

the options to tune a solid acid framework by organic doping are not extensive as regular pores are not typically observed. Layered solid acids^{37,38}, included in Fig. 6, will typically swell or exfoliate with hydration³⁹. Comparison is also made to sulfonated three-dimensional mesoporous solids such as PFSA-alumina⁴⁰. The mesopores in these solids enable covalent modification and hydrated samples to show good conductivity, but challenges with hydrogen fuel cross-over can certainly be envisaged. The MOF approach merges the options to alter the inorganic components of the solid acid approach with the choice of organic components from the macromolecular approach in an open, crystalline architecture that can be made gas-tight by guest loading. With exact knowledge of the dimensions of a pore and the locations of hydrogen bond donors and acceptors, the pairing with the ideal proton transferring dopant could be made judiciously with sub-angstrom level insight.

Research on MOFs has mainly focused on gas storage^{4,8}. Transport phenomena, including ion conduction, have been virtually ignored. To date, two families have been reported that shows ion conductivity. Cu thiooxamides that conduct protons in the 10^{-6} – $10^{-5} \text{ S cm}^{-1}$ range below 80°C with hydration dependence^{9,10}, and a recently reported ferrous oxalate hydrate that conducts on the order of $10^{-3} \text{ S cm}^{-1}$ at ambient conditions and 98% relative humidity¹¹. One MOF has been reported with sulfonated pores but no conductivity studies were performed⁴¹. The extrinsic doping of a MOF to give proton conduction above 100°C and its incorporation into a MEA to show a significant OCV are major advances. More generally, there are very few examples where an intrapore guest plays any role in the function of a MOF^{42,43}. The guests may act as substrates for catalysis^{44–47} or sensing^{48,49}, but typically they are simply removed to activate the pore for storage. This work represents a case where, not only is the nature of the guest fundamental to function, but the stoichiometry of loading is a determinant of performance.

We have presented a new approach to high-temperature proton conducting materials based on metal organic frameworks. PCMOFs can be designed based on molecular-level insights to

structure and conduction pathways. The selection of host and guest components for PCMOFs is vast and improvements can be made based on lessons learned from both macromolecular and solid acid approaches to proton conduction. Further, PCMOFs could be used, as inorganic salts have, with complementary polymer supports to form proton conducting nanocomposites⁵⁰. Ongoing work in our lab includes more robust PCMOFs with higher proton conductivity, optimal guest matching and mechanistic studies.

Methods

Procedure for the synthesis of β -PCMOF2. Anhydrous phloroglucinol (8.053 g, 63.86 mmol) was dissolved in dry dimethyl carbonate (200 ml) and the resulting yellow solution was cooled to 0°C under argon. Chlorosulfonic acid (12.43 ml, 186.8 mmol, 2.925 equiv.) was added dropwise over the course of two minutes with vigorous stirring, resulting in an orange coloured solution. The reaction was warmed to ambient temperature and allowed to proceed for 1.5 h under a steady flow of Ar before the system was sealed and stirred for an additional 22.5 h. The solvent was then removed via rotary evaporation and the resulting viscous brown oil was dissolved in 150 ml of water. Sodium bicarbonate was added to the solution until it reached a pH of 2.0. Addition of the aqueous solution to 4 l of acetone resulted in the formation of a thick white precipitate, which was isolated via vacuum filtration. This phase (α) was then converted to the β -phase via heating in a bomb apparatus. As synthesized α -PCMOF2 (0.500 g, 1.16 mmol) was placed into a 20 ml teflon cup. To this, a mixture of water (5 ml), methanol (1.5 ml) and acetone (1.5 ml) was added. The teflon cup was then sealed in a hydrothermal bomb. The bomb was heated at 100°C for 24 h and then cooled back to room temperature over a period of 12 h. The resulting white precipitate was isolated via vacuum filtration.

General procedure for the synthesis of β -PCMOF2(Tz)_{0.3}. Anhydrous phloroglucinol (6.150 g, 48.77 mmol) was dissolved in dry dimethyl carbonate (175 ml) and the resulting yellow solution cooled to 0°C under Ar. Chlorosulfonic acid (9.50 ml, 143 mmol, 2.93 equiv.) was added dropwise over the course of two minutes with vigorous stirring, resulting in an orange coloured solution. The reaction was warmed to ambient temperature and allowed to proceed for 1.5 h under a steady flow of Ar before the system was sealed and stirred for an additional 22.5 h. The solvent was then removed via rotary evaporation and the resulting viscous brown oil dissolved in water to a concentration of 125 mg ml^{-1} . 1,2,4-Triazole (3.283 g, 47.54 mmol), was then added to the aqueous solution and allowed to stir for 24 h. Sodium bicarbonate (5.992 g, 71.33 mmol, 1.5 equiv.) was added to the solution, which was then precipitated via addition to 3 l of acetone. The resulting thick white precipitate was isolated via vacuum filtration. Similarly, α -PCMOF2(Tz)_{0.45} was prepared by adding 1.35 equivalents (relative to ligand) of NaHCO_3 , and α -PCMOF2(Tz)_{0.6} was prepared by adding 1.2 equivalents (relative to ligand) of NaHCO_3 , during preparation of the frameworks. The α -phases were then quantitatively converted to β -phase in a bomb apparatus. Approximately 2 g of as synthesized α -PCMOF2(Tz) was packed into a silicone tube, sealed and pressed under a hydrostatic pressure of 5,000 kg for 2 min. The resulting pellet was removed from the silicone tube and placed into a 20 ml teflon cup, along with a vial containing 0.5 ml of water. The teflon cup was then sealed in a hydrothermal bomb and heated at 80°C for 8 h.

Impedance spectroscopy. Impedance analysis was performed on pristine PCMOF powders without modification. All pellets were prepared by first grinding the samples into a homogeneous powder with a mortar and pestle. The powders ($\sim 0.3 \text{ g}$) were then added to a standard 13 mm die, sandwiched between two porous carbon electrodes (Sigracet, GDL 10 BB, 1.222 cm diameter) and pressed at 5,000 kg for 2 min. All pellets were 13 mm in diameter and ranged in thickness from 1.30 to 1.38 mm. Measurements were performed using an impedance and gain-phase analyser (SI model 1260), 1 Hz–1 MHz, with a two-probe electrochemical cell equipped with platinum current collectors and an applied a.c. voltage of 10–100 mV (unless otherwise stated). Measurements were taken in the temperature range of 23 to 150°C in either an anhydrous H_2 or D_2 atmosphere. Conductivity was calculated using the following equation: $\sigma = l/R_s S$, where l and S are the thickness (cm) and cross-sectional area (cm^2) of the pellet respectively, and R_s , which was extracted directly from the impedance plots, is the bulk resistance of the sample (Ω).

Full experimental details and characterization are provided as Supplementary Information.

Received 20 July 2009; accepted 9 September 2009;
published online 18 October 2009

References

- Kitagawa, S. & Matsuda, R. Chemistry of coordination space of porous coordination polymers. *Coord. Chem. Rev.* **251**, 2490–2509 (2007).
- Yaghi, O. M. *et al.* Reticular synthesis and the design of new materials. *Nature* **423**, 705–714 (2003).

3. Mueller, U. *et al.* Metal-organic frameworks- prospective industrial applications. *J. Mater. Chem.* **16**, 626–636 (2006).
4. Collins, D. J. & Zhou, H. C. Hydrogen storage in metal-organic frameworks. *J. Mater. Chem.* **17**, 3154–3160 (2007).
5. Choi, H. J., Dinca, M. & Long, J. R. Broadly hysteretic H₂ adsorption in the microporous metal-organic framework Co(1,4-benzenedipyrazolate). *J. Am. Chem. Soc.* **130**, 7848–7850 (2008).
6. Wang, B., Côté, A. P., Furukawa, H., O'Keeffe, M. & Yaghi, O. M. Colossal cages in zeolitic imidazolate frameworks as selective carbon dioxide reservoirs. *Nature* **453**, 207–211 (2008).
7. Vaidhyanathan, R., Iremonger, S. S., Dawson, K. W. & Shimizu, G. K. H. An amine-functionalized metal organic framework for preferential CO₂ adsorption at low pressures. *Chem. Commun.* 5230–5232 (2009).
8. Ma, S. Q. *et al.* Metal-organic framework from an anthracene derivative containing nanoscopic cages exhibiting high methane uptake. *J. Am. Chem. Soc.* **130**, 1012–1016 (2008).
9. Nagao, Y., Fujishima, M., Ikeda, R., Kanda, S. & Kitagawa, H. Highly proton-conductive copper coordination polymers. *Synth. Metals* **133**, 431–432 (2003).
10. Nagao, Y. *et al.* Preparation and proton transport property of N,N'-diethyldithiooxamidatocopper coordination polymer. *Synth. Metals* **154**, 89–92 (2005).
11. Yamada, T., Sadakiyo, M. & Kitagawa, H. High proton conductivity of one-dimensional ferrous oxalate dehydrate. *J. Am. Chem. Soc.* **131**, 3144–3145 (2009).
12. Serre, C. *et al.* Role of solvent-host interactions that lead to very large swelling of hybrid frameworks. *Science* **315**, 1828–1831 (2007).
13. Choi, H. J. & Suh, M. P. Dynamic and redox active pillared bilayer open framework: single-crystal-to-single-crystal transformations upon guest removal, guest exchange, and framework oxidation. *J. Am. Chem. Soc.* **126**, 15844–15851 (2004).
14. Bradshaw, D., Warren, J. E. & Rosseinsky, M. J. Reversible concerted ligand substitution at alternating metal sites in an extended solid. *Science* **315**, 977–980 (2007).
15. Chandler, B. D. *et al.* Mechanical gas capture and release in a network solid via multiple single crystalline transformations. *Nature Mater.* **7**, 229 (2008).
16. Shimizu, G. K. H. Assembly of metal ions and ligands with adaptable coordinative tendencies as a route to functional metal-organic solids. *J. Solid State Chem.* **178**, 2519 (2005).
17. Zhang, J. *et al.* High temperature PEM fuel cells. *J. Power Sources* **160**, 872–891 (2006).
18. Mauritz, K. A. & Moore, R. B. State of understanding of Nafion. *Chem. Rev.* **104**, 4535–4586 (2004).
19. Kreuer, K. D., Paddison, S. J., Spohr, E. & Schuster, M. Transport in proton conductors for fuel-cell applications: simulations, elementary reactions, and phenomenology. *Chem. Rev.* **104**, 4637–4678 (2004).
20. Li, Q. F., He, R. H., Jensen, J. O. & Bjerrum, N. J. Approaches and recent development of polymer electrolyte membranes for fuel cells operating above 100 °C. *Chem. Mater.* **15**, 4896–4915 (2003).
21. Shimizu, G. K. H., Hurd, J. A., Vaidhyanathan, R. & Taylor, J. M. Metal-organic solids for use in proton exchange membranes. US Patent, 2008/0160356 A1 (2008).
22. Hickner, M. A., Ghassemi, H., Kim, Y. S., Einsla, B. R. & McGrath, J. E. Alternative polymer systems for proton exchange membranes (PEMs). *Chem. Rev.* **104**, 4587–4612 (2004).
23. Alberti, G. & Casciola, M. Solid state protonic conductors, present main applications and future prospects. *Solid State Ionics* **145**, 3–16 (2001).
24. Kreuer, K. D. Proton conductivity: materials and applications. *Chem. Mater.* **8**, 610–641 (1996).
25. Goward, G. R., Schuster, M. F. H., Sebastiani, D., Schnell, I. & Spiess, H. W. High-resolution solid-state NMR studies of imidazole-based proton conductors: Structure motifs and chemical exchange from ¹H NMR. *J. Phys. Chem. B* **106**, 9322–9334 (2002).
26. Li, S., Zhou, Z., Zhang, Y., Liu, M. & Li, W. 1H-1,2,4-Triazole: An effective solvent for proton-conducting electrolytes. *Chem. Mater.* **17**, 5884–5886 (2005).
27. Marti-Rujas, J., Desmedt, A., Harris, K. D. M. & Guillaume, F. Direct time-resolved and spatially resolved monitoring of molecular transport in a crystalline nanochannel system. *J. Am. Chem. Soc.* **126**, 11124–11125 (2004).
28. Irvine, J. T. S., Sinclair, D. C. & West, A. R. Electroceramics: Characterization by impedance spectroscopy. *Adv. Mater.* **2**, 132–138 (1990).
29. Casciola, M., Alberti, G., Sganappa, M. & Narducci, R. On the decay of Nafion proton conductivity at high temperature and relative humidity. *J. Power Sources* **162**, 141–145 (2006).
30. Haile, S. M., Chisholm, C. R. I., Sasaki, K., Boysen, D. A. & Uda, T. Solid acid proton conductors: from laboratory curiosities to fuel cell electrolytes. *Faraday Discuss.* **134**, 17–39 (2007).
31. Yamada, M. & Honma, I. Heteropolyacid-encapsulated self-assembled materials for anhydrous proton-conducting electrolytes. *J. Phys. Chem. B* **110**, 20486–20490 (2006).
32. Yamada, M. & Honma, I. An anhydrous proton conductor based on lactam-lactim tautomerism of uracil. *ChemPhysChem* **5**, 724–728 (2004).
33. Bhella, S. S. & Thangadurai, V. Synthesis and characterization of carbon dioxide and boiling water stable proton conducting double perovskite-type metal oxides. *J. Power Sources* **186**, 311–319 (2009).
34. Haile, S. Fuel cell materials and components. *Acta Materialia* **51**, 5981–6000 (2003).
35. Zhang, J. L., Tang, Y. H., Song, C. J., Zhang, J. J. & Wang, H. J. PEM fuel cell open circuit voltage (OCV) in the temperature range of 23 degrees C to 120 degrees C. *J. Power Sources*, **163**, 532–537 (2006).
36. Haile, S. M., Boysen, D. A., Chisholm, C. R. I. & Merle, R. B. Solid acids as fuel cell electrolytes. *Nature* **410**, 910–913 (2001).
37. Alberti, G., Boccali, L., Casciola, M., Massinelli, L. & Montoner, E. Protonic conductivity of layered zirconium phosphonates containing -SO₃H groups. 3. Preparation and characterization of gamma-zirconium sulfoaryl phosphonates. *Solid State Ionics* **84**, 97–104 (1996).
38. Casciola, M., Marmottini, F. & Peraio, A. AC conductivity of α-layered Zr phosphate in the presence of water vapor at 100–200 °C. *Solid State Ionics* **61**, 125–129 (1993).
39. Stein, E. W., Clearfield, A. & Subramanian, M. A. Conductivity of group IV metal sulphophosphonates and a new class of interstratified metal amine-sulphophosphonates. *Solid State Ionics* **83**, 113–124 (1996).
40. Athens, G. L., Ein-Eli, Y. & Chmelka, B. F. Acid-functionalized mesostructured aluminosilica for hydrophilic proton conduction membranes. *Adv. Mater.* **19**, 2580–2587 (2007).
41. Horike, S. *et al.* Immobilization of sodium ions on the pore surface of a porous coordination polymer. *J. Am. Chem. Soc.* **128**, 4222–4223 (2006).
42. Alkordi, M. H., Liu, Y. L., Larsen, R. W., Eubank, J. F. & Eddaoudi, M. Zeolite-like metal-organic frameworks as platforms for applications: On metalloporphyrin-based catalysts. *J. Am. Chem. Soc.* **130**, 12639–12641 (2008).
43. Hermes, S., Schroder, F., Amirjalayer, S., Schmid, R. & Fischer, R. A. Loading of porous metal-organic open frameworks with organometallic CVD precursors: inclusion compounds of the type [LnM]_n@MOF-5. *J. Mater. Chem.* **16**, 2464–2472 (2006).
44. Ohmori, O. & Fujita, M. Heterogeneous catalysis of a coordination network: cyanosilylation of imines catalyzed by a Cd(II)-(4,4'-bipyridine) square grid complex. *Chem. Commun.* 1586–1587 (2004).
45. Dybtsev, D. N. *et al.* Homochiral metal-organic material with permanent porosity, enantioselective sorption properties, and catalytic activity. *Angew. Chem. Int. Ed.* **45**, 916–920 (2006).
46. Cho, S. H., Ma, B., Nguyen, S. B. T., Hupp, J. T. & Albrecht-Schmitt, T. E. A metal-organic framework material that functions as an enantioselective catalyst for olefin epoxidation. *Chem. Commun.* 2563–2565 (2006).
47. Wu, C. D. & Lin, W. B. Heterogeneous asymmetric catalysis with homochiral metal-organic frameworks: Network-structure-dependent catalytic activity. *Angew. Chem. Int. Ed.* **46**, 1075–1078 (2007).
48. Neville, S. M. *et al.* Single-crystal to single-crystal structural transformation and photomagnetic properties of a porous iron(II) spin-crossover framework. *J. Am. Chem. Soc.* **130**, 2869–2876 (2008).
49. Katz, M. J., Ramnial, T., Yu, H. Z. & Leznoff, D. B. Polymorphism of Zn[Au(CN)₂]₂ and its luminescent sensory response to NH₃ vapor. *J. Am. Chem. Soc.* **130**, 10662–10673 (2008).
50. Alberti, G. & Casciola, M. Composite membranes for medium-temperature PEM fuel cells. *Annu. Rev. Mater. Sci.* **33**, 129–154 (2003).

Acknowledgements

The authors thank the Western Canada Fuel Cell Initiative, Alberta Energy Research Institute, University Technologies International, Southern Alberta Intellectual Property Alliance, the Natural Sciences and Engineering Research Council of Canada and the Canadian Foundation for Innovation for funding.

Author contributions

J.A.H. designed experiments, performed synthesis, characterization and property studies. R.V. grew crystals, performed crystallography and SEM/EDX. V.T. designed impedance and membrane electrode assembly experiments. C.I.R. and I.L.M. designed and performed solid state NMR experiments. G.K.H.S. conceived the project and drafted the manuscript. All authors discussed results and commented on the manuscript.

Additional information

Supplementary information accompanies this paper at www.nature.com/naturechemistry. Reprints and permission information is available online at <http://npg.nature.com/reprintsandpermissions/>. Correspondence and requests for materials should be addressed to G.K.H.S.